MODULE 03 CONTRIBUTION JOURNAL

ML WORKFLOW AND LEARNING TYPES

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MY ASSIGNED WORK

For the Module 03 group lab, I completed my individual review of the full 25-cell notebook and prepared my material for the group's final submission. I followed the notebook in order, ran the required code cells, checked their outputs, and completed the feature experiment, learning-type assessment, notebook reflection, and journal entries.

DATA AND VISUALIZATION CHECKS

I checked the Wine dataset and confirmed that it contained 178 samples, 13 input features, three target classes, and no missing values. I reviewed how the target labels and readable class names were added to the DataFrame. I also examined the class-distribution chart and correlation heatmap and noted their main patterns.

I verified the creation of the feature matrix X and target vector y. I also checked the 80/20 data split, which produced 142 training samples and 36 testing samples, and confirmed that the class proportions were maintained in both sets.

MODEL COMPARISON AND INDIVIDUAL EXPERIMENT

I ran the Logistic Regression and Decision Tree models and compared their evaluation results. I recorded that Logistic Regression achieved 88.9% accuracy and Decision Tree achieved 83.3%, making Logistic Regression the better original model. I reviewed the classification reports and checked the confusion matrix to identify correct predictions and misclassifications.

For the hands-on task, I replaced the original example features with alcohol, hue, and proline. I retrained Logistic Regression, compared the new accuracy with the original result, and documented that the selected features improved performance on the current test set. I kept the comparison in the notebook as evidence of my experiment.

ASSESSMENT, DOCUMENTATION, AND FINAL REVIEW

I completed the five-scenario learning-type assessment and verified a score of 5 out of 5. I wrote my own Reflection and Analysis responses, including an education application, the most challenging part of the lab, and ideas for improvement. I added observations where needed so that my work showed understanding rather than only code execution.

Finally, I reviewed the notebook from beginning to end, checked that the required cells, outputs, plots, and written responses were complete, and prepared my individual entries for the group's combined Reflection and Contribution documents.